

Course Number	0418-347	Course Title	Compiler Design
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Thalaya AlFozan
Compulsory	No	Prerequisite	0418-211

Textbook

<i>Title:</i>	<i>Compilers, Principles, Techniques, and Tools,</i>
<i>Author:</i>	<i>A. Aho, M. Lam, R. Sethi and J. Ullman</i>
<i>Publisher:</i>	<i>Addison-Wesley, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Fundamental structure of programming systems: Compilation and interpretation; T-diagram and bootstrapping; source-code, object-code, target-code; linking, loading; translation phases, compiler structure, interpreter structure; the concept of compiler writing systems.	6
2	Lexical analyzer - scanner construction: Regular sets and finite automata; transition diagrams and scanner, automaton table driven scanner.	8
3	Syntax analyzer - parser construction: Context-free grammars and languages; ambiguous and unambiguous grammars, equivalence of grammars; top down parsing, eliminating left recursion, left factoring, recursive descent parsing, backtracking, predictive parsing, LL(k)-grammars, LL(1)-control table, table-driven LL(1) parser; bottom up parsing, LR(k)-grammars, SLR(1) parser.	12
4	Semantic analysis and synthesis: Translation of declarations and symbol-table organization; L-attributed grammars and program translation; intermediate code generation.	8
5	Target program generation: Issues of the design of a target code generator.	5
6	Review & Exams.	3
	Total.	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
15-20%	Project
15-0%	Hmk/ quiz
40%	Final

Course Outcomes

Students should be able to:

1. Understand concepts and components of compilers and interpreters.
2. Apply the theory of formal languages and automata to describe the syntax of a programming language.
3. Understand and apply different parsing techniques.
4. Design and implement lexical and semantic analyzers.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	H					
2		H				
3		H				
4		H				